$$
\begin{aligned}
|\Psi(t)\rangle &= \hat{V}(t)\,|\Psi(0)\rangle \\
&= a\,|0\rangle \prod_i \hat{D}\left(\lambda_i(t)/2\right)|\Phi_{\mathcal{E}}\rangle + b\,|1\rangle \prod_i \hat{D}\left(-\lambda_i(t)/2\right)|\Phi_{\mathcal{E}}\rangle \\
&\equiv a\,|0\rangle\,|\mathcal{E}_+(t)\rangle + b\,|1\rangle\,|\mathcal{E}_-(t)\rangle\,. \qquad (5.102)
\end{aligned}
$$